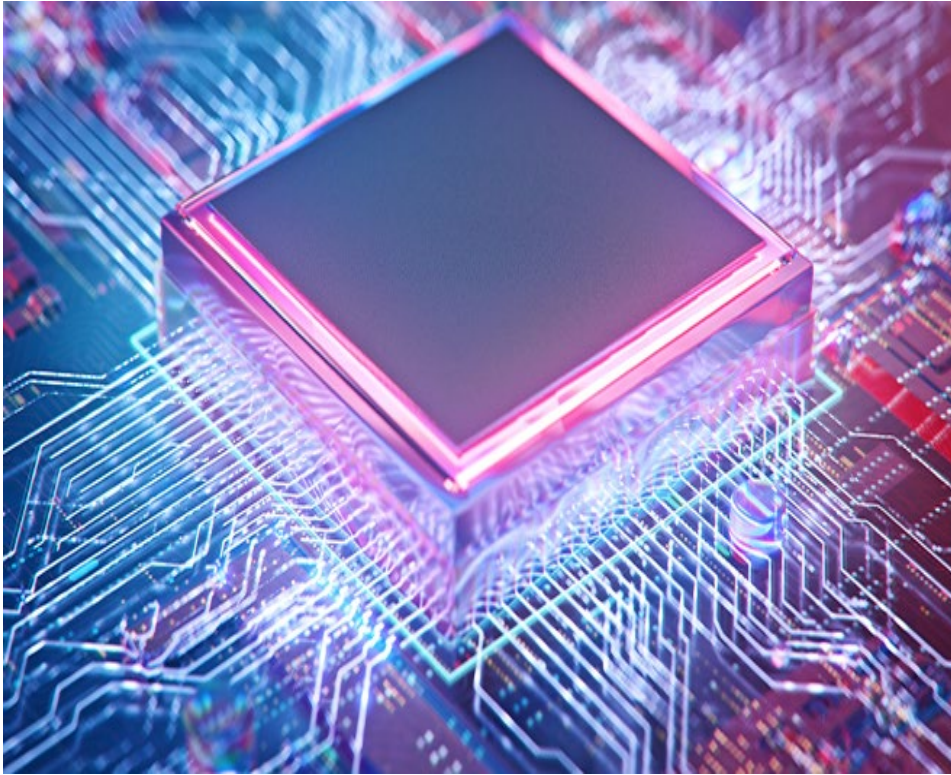


# Model 685 | 6 GHz 12.32 GS/s

High Performance Arbitrary Waveform Generator  
(Serial Pattern Generator and Digital Pattern Generator)



## Features

- 2, 4 or 8 Analog Channels
- 6.16 GS/s (12.32 GS/s in RF mode)
- 16-bit Vertical Resolution
- Up to 6 GHz Output Frequency
- < 110 ps Rise/Fall Time
- Up to 5 Vpp Output Voltage and  $\pm 2.5V$  Hardware Offset into 50  $\Omega$
- Up to 4 Gpts Waveform Memory per Channel
- Up to 32 Digital Channels in Synchronous with Analog Generation

## Applications

- Aerospace and Defense
- Institute and University Research
- Semiconductor Tests
- Automotive
- IoT



**Model 685 | 6 GHz 12.32 GS/s**  
Arbitrary Waveform Generator

# Model 685 Arbitrary Waveform Generator

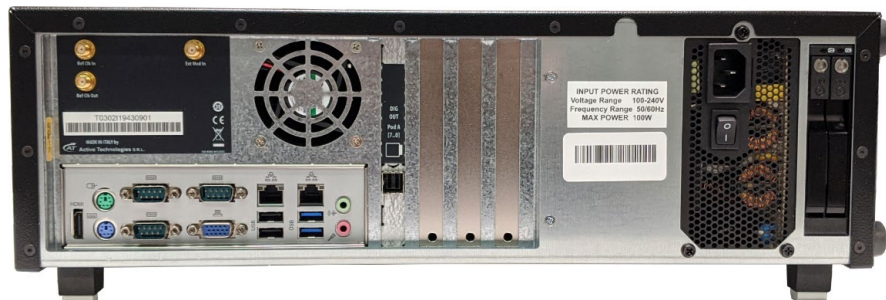
## Features & Benefits

- Sample rate can be programmed in from 1 S/s to 6.16 GS/s (12.32 GS/s in RF mode), with 16-bit vertical resolution, ensures exceptional signal integrity.
- Arbitrary waveform memory up to 4 Gpts for each analog channel
- Mixed Signal Generation – 2, 4, or 8 Analog channels with 8, 16, or 32 synchronized Digital Channels for debugging and validating digital design.
- Three operation modes – Simple Rider AFG (DDS AFG mode), True Arb (variable clock Arbitrary AWG mode) and SPG (Serial Pattern Generator).
- Digital outputs provide up to 1.54 Gb/s data rate in LVDS format. LVDS to LVTTTL adapter is available
- Advance sequencer with up to 16384 user-defined waveforms provides the possibility of generating complex signal scenarios with the most efficient memory usage
- Windows-based platform with 7in touchscreen, front panel buttons and knob
- Compact form factor, convenient for bench top and fully fit with 3U – 19" rack-mount standard
- LAN,USB-TMC and GPIB interfaces for remote control

## Model 685 Front



## Model 685 Back





# Model 685 Arbitrary Waveform Generator

## Model 685 User Interface

### Simple Rider AFG: Function Generator Mode Interface

**Simple Rider AFG** UI is designed using a touchscreen interface; it has been developed to put all the capabilities of modern Waveform Generators right at your fingertips. All instrument controls and parameters are accessible through an intuitive UI that recalls the simplicity of modern tablets and smartphones. Scientists and engineers can use gestures and touch commands to create advanced waveforms or digital patterns with ease.

- The swipe gesture gives easy access to the output waveform parameters.
- A touch-friendly virtual numeric keypad has been designed to improve the user experience with regard to entering data.
- Time-saving shortcuts and intuitive icons simplify the instrument setup.



### Simple Rider TrueArb: AWG and DPG Mode Interface

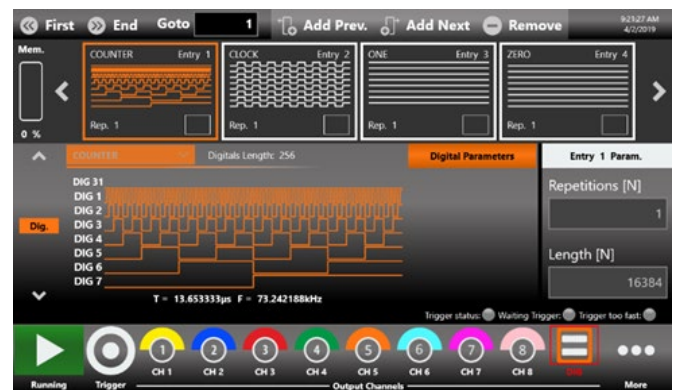
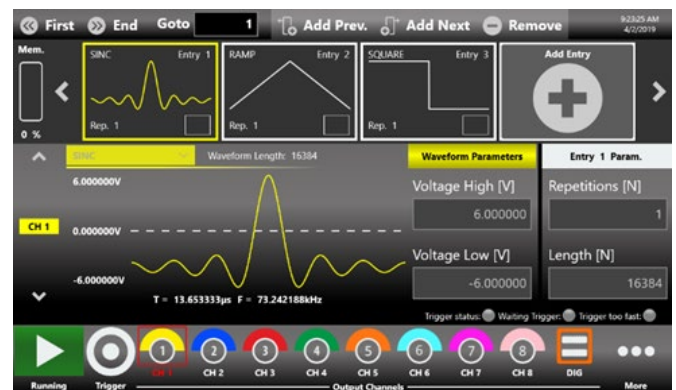
In the **Simple Rider True-Arb** interface, the users can define complex waveforms with up to 16,384 sequence entries of analog waveforms and digital patterns, define their execution flow by means of loops, jumps, and conditional branches.

Digital output combined and synchronized with analog output signals is an ideal tool to troubleshoot and validate the digital design. The waveform memory length of up to 4 GSamples on each channel combined with up to 16,384 and up to 4,294,967,294 repetitions, make the Model 685 the ideal generator for the most demanding technical applications.

Thanks to the intuitive and easy waveform sequencer user interface, you can create the most complex waveform scenarios with just a few screen touches.

Up to 4 instruments can be synchronized together in order to obtain a 32 analog – 128 digital channel generator. A dedicated synchronization bus guarantees intra-chassis synchronization.

Model 685 supports the standard Ethernet interface for remote control and easy customized instrument programming.

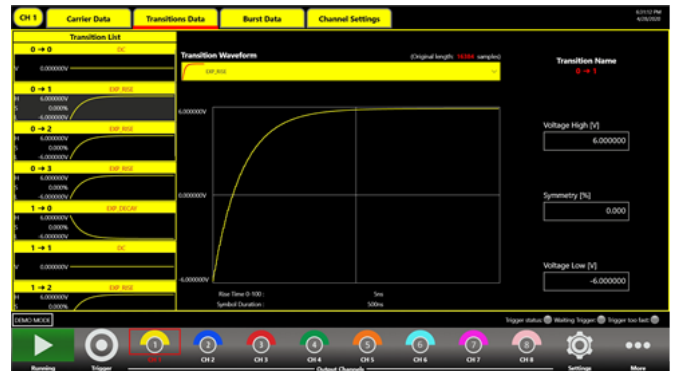


# Model 685 Arbitrary Waveform Generator

## Simple Rider SPG: Serial Pattern Generator (SPG) Mode Interface

The easiest touchscreen display interface allows creating patterns scenarios, only in a few screen touches.

In summary, the Data Pattern Generator provides the capability to generate PRBS patterns and up to 2MSymbols custom patterns where bit transitions can have arbitrarily user-defined shapes. The Model 685 Serial Pattern Generator can generate patterns up to 1.5Gbaud.



The software architecture provides the possibility to easily generate the patterns in different generation modalities and also gives the opportunity to modulate the patterns with internal or external signals with the purpose to generate also different effects of noise (jitter, ripple, ...).

## Model 685 Applications Automotive

Today's cars include highly sophisticated and electronically controlled units with sensitive electronic components. The Model 685 combines a 6.16 GSa/s sampling rate with a 16-bit vertical resolution. It represents an ideal and a successful tool for addressing the new testing challenges in the automotive industry. It includes these features:

- EMI debugging, troubleshooting, and testing
- Electrical standards emulation up to 5V



## IoT and Ind 4.0 Perfect RF Modulator

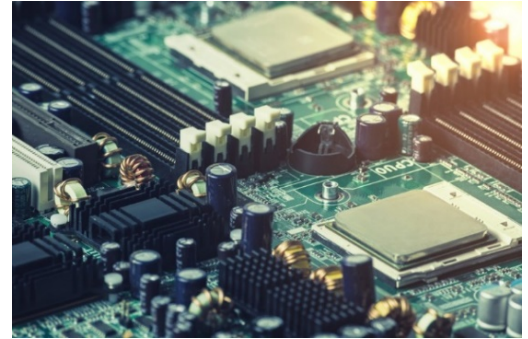
The Model 685 is the iconic instrument for this application. The possibility to emulate complex RF I/Q modulation for simulation and test vs wireless devices or working on the Internet of Things of Industry 4.0 applications. Each engineer may import waveforms to emulate devices for test purposes and impose distortion on waveforms (such as noise) to test device compliance with standards.



# Model 685 Arbitrary Waveform Generator

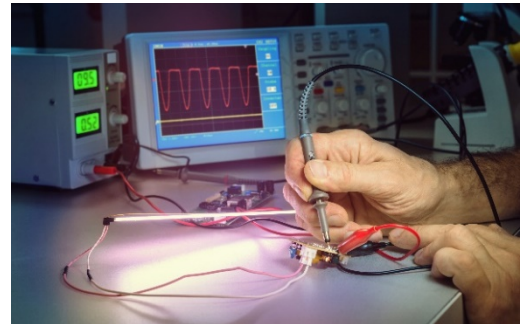
## Semiconductor Testing

Emulation of complex signals generated with inclusion of noise or distortions may become an excellent way to provide Compliance Components Test to help semiconductor engineers. The fast edges and pulse generation can be used to provide characterization in fast power devices.



## Research Applications

Research centers and Universities are key users of the Model 685. Complex waveform and/or sophisticated Pulses emulation based on variable edges or multilevel could be perfectly created. The combination of fast edge generation, excellent dynamic range and easy-to-use user interface meet perfectly for scientists and engineers working on Quantum Research or on large experiments such as Accelerators, Tokamak, or synchrotrons to emulate signals without creating specific test boards.



- Emulation of detectors
- Emulation of signal sources adding noise
- Generation/playback of real-world signals
- Emulation of long PRBS sequences
- Modulating and driving laser diode

## Aerospace and Defense applications

The Model 685 works perfectly with electronic warfare signals, such as those produced by radar or sonar systems. This generator can also be fitted into a modular system for radio or I/Q signal modulation. It can create pulses useful in pulse electron beams, X-ray sources, flash X-ray radiography, lightning pulse simulators, and high-power microwave modulators. These features of Model 685 are useful in the aerospace and defense industry.



- Frequency response, intermodulation distortion and noise-figure measurements
- Phase Locked Loop (PLL) pull-in and hold range characterization
- Radar base-band signals emulation

# Model 685 Arbitrary Waveform Generator

## Model 685 Specifications

All specifications are typical unless noted otherwise. The guaranteed performances are referred to a calibrated instrument that has been stored for a minimum of 2 hours within the operating temperature range of 5°C to 40°C and after a 45-minute warm-up period. Within  $\pm 10^{\circ}\text{C}$  after auto-calibration.

| General Specifications                                                                 |                                              |                               |                                                      |
|----------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------|------------------------------------------------------|
| Operating Mode                                                                         | AFG Mode<br>True Arb Mode<br>SPG Mode        |                               |                                                      |
| Number of Channels                                                                     | Model 685-2C<br>Model 685-2CD                | Model 685-4C<br>Model 685-4CD | Model 685-8C<br>Model 685-8CD                        |
| Analog                                                                                 | 2                                            | 4                             | 8                                                    |
| Digital Out                                                                            | 0/8 optional                                 | 0/8/16 optional               | 0/8/16/24/32 optional                                |
| Marker Out                                                                             | 1                                            | 2                             | 4                                                    |
|                                                                                        | Model 685-2C<br>Model 685-4C<br>Model 685-8C |                               | Model 685-2CD<br>Model 685-4CD<br>Model 685-8CD      |
| Output Channels                                                                        |                                              |                               |                                                      |
| Output Type                                                                            | Single ended DC coupled                      |                               | Differential DC coupled                              |
| Output Impedance                                                                       | Single-ended: 50 Ω                           |                               | Single ended: 50 Ω<br>Differential: 100 Ω            |
| Connectors                                                                             | SMA on front panel                           |                               |                                                      |
| DC Amplitude                                                                           |                                              |                               |                                                      |
| Amplitude Range                                                                        | ±2.5 V (into 50 Ω)                           |                               | ±0.75 V Se. (into 50 Ω)<br>±1.5 V Diff. (into 100 Ω) |
| Resolution                                                                             | 100μV (nom), 5 digits                        |                               |                                                      |
| Amplitude Accuracy (guaranteed)                                                        | ±(1% of  setting  + 5 mV)                    |                               | ±(0.5% of  setting  + 2 mV)                          |
| DC Baseline Hardware Offset (Common mode offset)                                       |                                              |                               |                                                      |
| Resolution                                                                             | < 4 mV or 4 digits                           |                               |                                                      |
| Range (50 Ω into 50 Ω)                                                                 | -2.5 V to +2.5 V                             |                               | -2 V to +2 V                                         |
| Range (50 Ω into High Z load)                                                          | -2.5 V to +2.5 V                             |                               | -4 V to +4 V                                         |
| Accuracy (50 Ω into 50 Ω) (guaranteed)                                                 | ± (1% of  setting  + 5 mV)                   |                               |                                                      |
| AC Accuracy (1 kHz sine wave, 0 V offset, > 5 mVp-p amplitude, 50 Ω load) (guaranteed) | ± (1% of setting [Vpp] + 5mV)                |                               |                                                      |

| True Arb - Baseband Mode Specifications | Model 685-2C<br>Model 685-4C<br>Model 685-8C                                                      | Model 685-2CD<br>Model 685-4CD<br>Model 685-8CD |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------|
| General specifications                  |                                                                                                   |                                                 |
| Operating Mode                          | Variable clock (True Arbitrary) – Baseband mode                                                   |                                                 |
| Sample Rate                             | 1 S/s to 6.16 GS/s                                                                                |                                                 |
| Sin(x)/x                                | 2.72 GHz @ 6.16GS/S,                                                                              |                                                 |
| Run Modes                               | Continuous, Triggered Continuous, Single/Burst, Stepped, Advanced                                 |                                                 |
| Vertical Resolution                     | 16 bit                                                                                            |                                                 |
| Waveform Length                         | 128 to 2G samples per channel (up to 4G samples optional)                                         |                                                 |
| Waveform Granularity                    | 1 if the entry length is > 416 samples<br>32 if entry length is $\geq 128$ and $\leq 416$ samples |                                                 |
| Sequence Length                         | 1 to 16384                                                                                        |                                                 |
| Sequence Repeat Counter                 | 1 to 4294967294 or infinite                                                                       |                                                 |



# Model 685 Arbitrary Waveform Generator

|                                                                                                                                |                                                                                                                                  |                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timer<br>Range<br>Resolution                                                                                                   | 20 ns to 1.39 seconds<br>± 1 sampling clock cycle                                                                                |                                                                                                                                                                     |
| Analog Channel to Channels skew                                                                                                |                                                                                                                                  |                                                                                                                                                                     |
| Range                                                                                                                          | 0 to 2.63 us                                                                                                                     |                                                                                                                                                                     |
| Resolution                                                                                                                     | 100 fs                                                                                                                           |                                                                                                                                                                     |
| Accuracy                                                                                                                       | ±(1% of setting + 20 ps)                                                                                                         |                                                                                                                                                                     |
| Initial skew                                                                                                                   | < 20 ps                                                                                                                          |                                                                                                                                                                     |
| Calculated bandwidth (0.35 / rise or fall time)                                                                                | ≥ 2 GHz                                                                                                                          | ≥ 2.2 GHz                                                                                                                                                           |
| SFDR @ 100 MHz (Fsa= 6 Gsa/s, measured across DC to Fs/2, excluding fsa - 2*fout and fsa- 3*fout and excluding harmonic)       | < -80 dBc                                                                                                                        | < - 90 dBc                                                                                                                                                          |
| SFDR (Fsa= 6,16 Gsa/s, measured across DC to Fs/2, excluding fsa - 2*fout and fsa- 3*fout and excluding harmonic) <sup>1</sup> | 1 μHz to ≤ 600 MHz: < -80 dBc<br>600 MHz to ≤ 1.5 GHz: < -75 dBc<br>1.5 GHz to ≤ 2 GHz: < -65 dBc<br>2 GHz to ≤ 3 GHz: < -55 dBc | 1 μHz to < 100 MHz: < -90 dBc<br>100 MHz to ≤ 600 MHz: < -82 dBc<br>600 MHz to ≤ 1.5 GHz: < -75 dBc<br>1.5 GHz to ≤ 2 GHz: < -70 dBc<br>2 GHz to ≤ 3 GHz: < -62 dBc |
| Rise/fall time (1 Vp-p single-ended 10% to 90%)                                                                                | ≤ 175 ps                                                                                                                         | ≤ 155 ps                                                                                                                                                            |
| Rise/fall time (1 Vp-p single-ended 20% to 80%)                                                                                | ≤ 110 ps                                                                                                                         | ≤ 100 ps                                                                                                                                                            |
| Overshoot (1 Vp-p single-ended)                                                                                                | <5%                                                                                                                              | <6%                                                                                                                                                                 |
| Random jitter on clock pattern (rms, typical)                                                                                  | < 2 ps                                                                                                                           |                                                                                                                                                                     |

| True Arb - Baseband mode specifications | Model 685-2C<br>Model 685-4C<br>Model 685-8C                                                                  | Model 685-2CD<br>Model 685-4CD<br>Model 685-8CD |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>General specifications</b>           |                                                                                                               |                                                 |
| Operating Mode                          | Variable clock (True Arbitrary) – RF mode                                                                     |                                                 |
| Output Sample Rate                      | 8.5 GS/s to 12.32 GS/s                                                                                        |                                                 |
| Sin(x)/x                                | 5.04 GHz @ 12.32GS/S                                                                                          |                                                 |
| RF Modulation                           | I/Q quadrature                                                                                                |                                                 |
| RF Carrier Count per Output Channel     | Single Carrier (2 components I0,Q0 for channel)<br>Double Carrier (4 components, I0,Q0 and I1,Q1 for channel) |                                                 |
| RF Carrier Frequency Range              | 0 up to 6 GHz                                                                                                 |                                                 |
| RF Carrier Frequency Resolution         | 1 mHz                                                                                                         |                                                 |
| RF Carrier Phase                        | Programmable                                                                                                  |                                                 |
| I/Q Component Data Rate                 | 1/8 of the Output Sample rate                                                                                 |                                                 |
| I/Q Component Prescaler                 | 0 to 2 <sup>32</sup>                                                                                          |                                                 |
| Run Modes                               | Continuous, Triggered Continuous, Single/Burst, Stepped, Advanced                                             |                                                 |
| I/Q Component Vertical Resolution       | 16 bit                                                                                                        |                                                 |
| I/Q Component Waveform Length           | 32M to 500M samples for component (up to 1G samples optional)                                                 |                                                 |
| I/Q Component Waveform Granularity      | 1 if the entry length is > 104 samples<br>8 if entry length is ≥ 32 and ≤ 104 samples                         |                                                 |
| Sequence Length                         | 1 to 16384                                                                                                    |                                                 |
| Sequence Repeat Counter                 | 1 to 4294967294 or infinite                                                                                   |                                                 |
| Timer<br>Range<br>Resolution            | 20 ns to 1.39 seconds<br>± 1 Component sampling clock cycle                                                   |                                                 |

<sup>1</sup> For 685-2C/4C/8C models the SFDR is evaluated @ 2.5Vpp single ended nominal output amplitude. For 685-2CD/4CD/8CD models the SFDR is evaluated @ 1.5Vpp differential nominal output amplitude provided to the spectrum analyzer through a Minicircuit TC1-1-13M+ balun.

# Model 685 Arbitrary Waveform Generator

| I/Q Component to Component skew |                                          |
|---------------------------------|------------------------------------------|
| Range                           | 0 to [16200 * 8/Output Sampling Clock] s |
| Resolution                      | [8/Output Sampling Clock] s              |
| Accuracy                        | ±(1% of setting + 20 ps)                 |
| Initial Skew                    | < 20 ps                                  |

| AFG Mode Specifications                            | Model 685-2C<br>Model 685-4C<br>Model 685-8C                                                                              | Model 685-2CD<br>Model 685-4CD<br>Model 685-8CD                                                                                      |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| General Specifications                             |                                                                                                                           |                                                                                                                                      |
| Amplitude                                          | 0 to 5 Vpp (into 50 Ω)                                                                                                    | 0 to 3 Vpp Diff. (into 100 Ω)<br>0 to 1.5 Vpp Se. (into 50 Ω)                                                                        |
| Resolution                                         | 100 μV (nom), 5 digits                                                                                                    |                                                                                                                                      |
| Operating Mode                                     | DDS mode                                                                                                                  |                                                                                                                                      |
| Standard Waveforms                                 | Sine, Square, Pulse, Ramp, more (Noise, DC, Sin(x)/x, Gaussian, Lorentz, Exponential Rise, Exponential Decay, Haversine   |                                                                                                                                      |
| Run Modes                                          | Continuous, modulation, sweep, burst                                                                                      |                                                                                                                                      |
| Arbitrary Waveforms                                | Vertical resolution: 16-bit<br>Waveform length: 16,384 points                                                             |                                                                                                                                      |
| Internal Trigger Timer                             |                                                                                                                           |                                                                                                                                      |
| Range                                              | 10.4 ns to 88 s                                                                                                           |                                                                                                                                      |
| Resolution                                         | 80 ps                                                                                                                     |                                                                                                                                      |
| Accuracy                                           | ±(0.1% setting + 5 ps)                                                                                                    |                                                                                                                                      |
| Sine Waves                                         |                                                                                                                           |                                                                                                                                      |
| Frequency Range Sine (50 Ω into 50 Ω) <sup>2</sup> | 1 μHz to ≤ 1 GHz: 5Vpp<br>1 GHz to ≤ 2 GHz: 4Vpp                                                                          | 1 μHz to ≤ 2 GHz: 3 Vpp Diff.<br>1 μHz to ≤ 2 GHz: 1.5 Vpp Se.                                                                       |
| Flatness                                           | DC to 2 GHz: ±0.5 dB<br>(1 Vpp, relative to 1 kHz)                                                                        | DC to 2 GHz: ±0.5 dB<br>(1 Vpp diff., relative to 1 kHz)                                                                             |
| Harmonic Distortion (1 Vp-p)                       | 1 μHz to ≤ 20 kHz < -75 dBc<br>20 kHz to ≤ 400 MHz < -70 dBc<br>400 MHz to ≤ 1GHz < -60 dBc<br>1 GHz to ≤ 2 GHz < -55 dBc |                                                                                                                                      |
| Total Harmonic Distortion (1 Vp-p)                 | 10 Hz to 20 kHz < 0.05%                                                                                                   |                                                                                                                                      |
| Spurious (measured across DC to Fs/2) <sup>3</sup> | 1 μHz to ≤ 500 MHz: < -75 dBc<br>500 MHz to ≤ 1.5 GHz: < -70 dBc<br>1.5 GHz to ≤ 2 GHz: < -55 dBc                         | 1 μHz to ≤ 250 MHz: < -85 dBc<br>250 MHz to ≤ 500 MHz: < -80 dBc<br>500 MHz to ≤ 1.5 GHz: < -70 dBc<br>1.5 GHz to ≤ 2 GHz: < -60 dBc |
| Phase Noise (1 Vp-p, 10 kHz offset)                | 20 MHz: < -127 dBc/Hz typ.<br>100 MHz: < -123 dBc/Hz typ.<br>1 GHz: < -105 dBc/Hz typ.                                    |                                                                                                                                      |
| Square Waves                                       |                                                                                                                           |                                                                                                                                      |
| Frequency Range                                    | 1 μHz to ≤ 770 MHz                                                                                                        | 1 μHz to ≤ 770 MHz                                                                                                                   |
| Rise/fall Time (10% to 90%)                        | 400 ps                                                                                                                    |                                                                                                                                      |
| Rise/fall Time (20% to 80%)                        | 300 ps                                                                                                                    |                                                                                                                                      |
| Overshoot (1 Vp-p)                                 | <2%                                                                                                                       |                                                                                                                                      |
| Jitter (rms)                                       | <2 ps                                                                                                                     |                                                                                                                                      |
| Pulse Waves                                        |                                                                                                                           |                                                                                                                                      |
| Frequency Range                                    | 1 μHz to ≤ 770 MHz                                                                                                        | 1 μHz to ≤ 770 MHz                                                                                                                   |
| Pulse Width                                        | 500 ps to (Period – 500 ps) <sup>4</sup>                                                                                  |                                                                                                                                      |
| Pulse Width Resolution                             | 20 ps or 15 digits                                                                                                        |                                                                                                                                      |

<sup>2</sup> Amplitude doubles on HiZ load

<sup>3</sup> For 685-2C/4C/8C models the spurious are evaluated @ 1Vpp single ended nominal output amplitude. For 685-2CD/4CD/8CD models the SFDR is evaluated @ 1Vpp differential nominal output amplitude provided to the spectrum analyzer through a Minicircuit TC1-1-13M+ balun.

<sup>4</sup>. Below 500 ps width, the pulse amplitude will have some reduction respect to the set value.



# Model 685 Arbitrary Waveform Generator

|                                                                                                  |                                                                 |                                                                                                                       |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Pulse Duty                                                                                       | 0.1% to 99.9% (limitations of pulse width apply)                |                                                                                                                       |
| Leading/trailing edge transition time (10% to 90%)                                               | 400 ps to 1000 s                                                |                                                                                                                       |
| Leading/trailing edge transition time (20% to 80%)                                               | 300 ps to 1000 s                                                |                                                                                                                       |
| Transition Time Resolution                                                                       | 2 ps or 15 digits                                               |                                                                                                                       |
| Overshoot (1 Vp-p)                                                                               | < 2%                                                            |                                                                                                                       |
| Jitter (rms, with rise and fall time ≥ 400ps)                                                    | <2 ps                                                           |                                                                                                                       |
| Double Pulse Waves                                                                               |                                                                 |                                                                                                                       |
| Frequency Range                                                                                  | 1 μHz to ≤ 385 MHz: 10 Vpp<br>where<br>Vpp=  Vpp1  +  Vpp2      | 1 μHz to ≤ 385 MHz: 6 Vpp Diff.<br>(1 μHz to ≤ 385 MHz: 3 Vpp Se)<br>where<br>Vpp=  Vpp1  +  Vpp2                     |
| Other Pulse Parameters                                                                           | Same as Pulse Waves                                             |                                                                                                                       |
| Ramp Waves                                                                                       |                                                                 |                                                                                                                       |
| Frequency Range                                                                                  | 1 μHz to 75 MHz                                                 |                                                                                                                       |
| Linearity (< 10 kHz, 1 Vp-p, 100%)                                                               | ≤ 0.1%                                                          |                                                                                                                       |
| Symmetry                                                                                         | 0% to 100%                                                      |                                                                                                                       |
| Other Waves                                                                                      |                                                                 |                                                                                                                       |
| Frequency Range<br>Exponential Rise, Exponential Decay<br>Sin(x)/x, Gaussian, Lorentz, Haversine | 1 μHz to 75 MHz<br>1 μHz to 150 MHz                             |                                                                                                                       |
| Additive Noise                                                                                   |                                                                 |                                                                                                                       |
| Bandwidth (-3 dB)                                                                                | 2 GHz                                                           |                                                                                                                       |
| Level                                                                                            | 0 V to 2.5 V - abs(carrier max value [Vpk])                     | 0 V to 0.75 V Single Ended - abs(carrier max value [Vpk])<br>0 V to 1.5 V Differential - abs(carrier max value [Vpk]) |
| Resolution                                                                                       | 1 mV                                                            |                                                                                                                       |
| Arbitrary                                                                                        |                                                                 |                                                                                                                       |
| Number of Samples                                                                                | 2 to 16384                                                      |                                                                                                                       |
| Frequency Range                                                                                  | 1 μHz to ≤ 770 MHz                                              |                                                                                                                       |
| Analog Bandwidth (-3 dB)                                                                         | 950 MHz                                                         |                                                                                                                       |
| Rise/Fall Time (10% to 90%)                                                                      | 400 ps                                                          |                                                                                                                       |
| Rise/Fall Time (20% to 80%)                                                                      | 300 ps                                                          |                                                                                                                       |
| Jitter (rms)                                                                                     | < 2 ps                                                          |                                                                                                                       |
| Frequency Resolution                                                                             |                                                                 |                                                                                                                       |
| Sine, Square, Pulse, Arbitrary, Sin(x)/X                                                         | 1 μHz or 15 digits                                              |                                                                                                                       |
| Gaussian, Lorentz, Exponential Rise, Exponential Decay, Haversine                                | 1 μHz or 14 digits                                              |                                                                                                                       |
| Frequency Accuracy                                                                               |                                                                 |                                                                                                                       |
| Non-ARB                                                                                          | ± 2.0 ppm of setting   ± 500 ppb of setting (Opt.)              |                                                                                                                       |
| ARB                                                                                              | ± 2.0 ppm of setting ±1 μHz   ± 500 ppb of setting ±1 μHz(Opt.) |                                                                                                                       |
| Modulations                                                                                      |                                                                 |                                                                                                                       |
| Amplitude Modulation (AM)                                                                        |                                                                 |                                                                                                                       |
| Carrier Waveforms                                                                                | Standard waveforms (except Pulse, DC and Noise), ARB            |                                                                                                                       |
| Modulation Source                                                                                | Internal or external                                            |                                                                                                                       |
| Internal Modulating Waveforms                                                                    | Sine, Square, Ramp, Noise, ARB                                  |                                                                                                                       |
| Modulating Frequency                                                                             | Internal: 500 μHz to 61 MHz, External: 10 MHz maximum           |                                                                                                                       |
| Depth                                                                                            | 0.00% to 120.00%                                                |                                                                                                                       |
| Frequency Modulation (FM)                                                                        |                                                                 |                                                                                                                       |
| Carrier Waveforms                                                                                | Standard waveforms (except Pulse, DC and Noise), ARB            |                                                                                                                       |

# Model 685 Arbitrary Waveform Generator

|                                     |                                                            |
|-------------------------------------|------------------------------------------------------------|
| Modulation Source                   | Internal or external                                       |
| Internal Modulating Waveforms       | Sine, Square, Ramp, Noise, ARB                             |
| Modulating Frequency                | Internal: 500 $\mu$ Hz to 61 MHz, External: 10 MHz maximum |
| Peak Deviation                      | DC to 2 GHz                                                |
| <b>Phase Modulation (PM)</b>        |                                                            |
| Carrier Waveforms                   | Standard waveforms (except Pulse, DC and Noise), ARB       |
| Modulation Source                   | Internal or external                                       |
| Internal Modulating Waveforms       | Sine, Square, Ramp, Noise, ARB                             |
| Modulating Frequency                | Internal: 500 $\mu$ Hz to 61 MHz, External: 10 MHz maximum |
| Phase Deviation Range               | 0° to 360°                                                 |
| <b>Frequency Shift Keying (FSK)</b> |                                                            |
| Carrier Waveforms                   | Standard waveforms (except Pulse, DC and Noise), ARB       |
| Modulation Source                   | Internal or external                                       |
| Internal Modulating Waveforms       | Square                                                     |
| Key Rate                            | Internal: 500 $\mu$ Hz to 61 MHz, External: 10 MHz maximum |
| Hop Frequency                       | 1 $\mu$ Hz to 2 GHz                                        |
| Number of Keys                      | 2                                                          |
| <b>Phase Shift Keying (PSK)</b>     |                                                            |
| Carrier Waveforms                   | Standard waveforms (except Pulse, DC and Noise), ARB       |
| Modulation source                   | Internal or external                                       |
| Internal Modulating Waveforms       | Square                                                     |
| Key Rate                            | Internal: 500 $\mu$ Hz to 61 MHz, External: 10 MHz maximum |
| Hop Frequency                       | 0° to +360°                                                |
| Number of Keys                      | 2                                                          |
| <b>Pulse Width Modulation (PWM)</b> |                                                            |
| Carrier Waveforms                   | Pulse                                                      |
| Modulation Source                   | Internal or external                                       |
| Internal Modulating Waveforms       | Sine, Square, Ramp, Noise, ARB                             |
| Modulating Frequency                | Internal: 500 $\mu$ Hz to 61 MHz, External: 10 MHz maximum |
| Deviation Range                     | 0% to 50% of pulse period                                  |
| <b>Sweep</b>                        |                                                            |
| Type                                | Linear, Logarithmic, staircase, and user defined           |
| Waveforms                           | Standard waveforms (except Pulse, DC and Noise), ARB       |
| Sweep Time                          | 30 ns to 2000 s                                            |
| Hold/return Times                   | 0 to (2000 s - 30 ns)                                      |
| Sweep/Hold/Return Time Resolution   | 15 ns or 12 digits                                         |
| Total Sweep Time Accuracy           | $\leq 0.4\%$                                               |
| Start/Stop Frequency Range          | Sine: 1 $\mu$ Hz to 2 GHz, Square: 1 $\mu$ Hz to 770 MHz   |
| Trigger Source                      | Internal/External/Manual                                   |
| <b>Burst</b>                        |                                                            |
| Waveforms                           | Standard waveforms (except DC and Noise), ARB              |
| Type                                | Trigger or gated                                           |
| Burst Count                         | 1 to 4,294,967,295 cycles or Infinite                      |

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| <b>Timing and Clock</b> |                                                        |
| <b>Sampling Rate</b>    |                                                        |
| Range                   | 1 S/s to 6.16 GS/s<br>(1 S/s to 12.32 GS/S in RF mode) |
| Resolution              | 32 Hz                                                  |
| Accuracy                | $\pm 2.0$ ppm   $\pm 500$ ppb (Opt.)                   |

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| Digital outputs (Optional)                                      |                                                                                      |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Output Channels</b>                                          |                                                                                      |
| Connectors                                                      | Mini-SAS HD connector on rear panel (custom pin-out)                                 |
| Number of Connectors                                            | 1, 2, 4                                                                              |
| Number of Outputs                                               | 8-bits, 16-bits, 32-bits                                                             |
| <b>Output impedance</b>                                         | 100 $\Omega$ differential                                                            |
| <b>Output type</b>                                              | LVDS                                                                                 |
| <b>Rise/fall time</b> (10% to 90%)                              | < 1 ns                                                                               |
| <b>Jitter</b> (rms)                                             | 20 ps                                                                                |
| <b>Maximum update rate</b>                                      | 1.54 Gbps per channel                                                                |
| <b>Memory depth</b>                                             | 512M Samples per digital channel (up to 1G optional)                                 |
| <b>8 bit LVDS to LVTTTL Converter Probe (Optional AT-DTLL8)</b> |    |
| <b>Output Connector</b>                                         | 20 position 2.54 mm 2 Row IDC Header                                                 |
| <b>Output Type</b>                                              | LVTTTL                                                                               |
| <b>Output Impedance</b>                                         | 50 $\Omega$ nominal                                                                  |
| <b>Output Voltage</b>                                           | 0.8 V to 3.8 V programmable in group of 8 bits                                       |
| <b>Maximum Update Rate</b>                                      | 125 Mbps@0.8V and 400 Mbps@3.6V                                                      |
| <b>Dimensions</b>                                               | W 2in x H 0.9in x D 3in [52mm x 22mm x 76mm]                                         |
| <b>Input Connector</b>                                          | Proprietary standard                                                                 |
| <b>Cable Length</b>                                             | 1 meter                                                                              |
| <b>Cable Type</b>                                               | Proprietary standard                                                                 |
| <b>Proprietary Mini SAS HD to SMA cable (Optional)</b>          |  |
| <b>Output Connector</b>                                         | SMA                                                                                  |
| <b>Output Type</b>                                              | LVDS                                                                                 |
| <b>Number of SMA</b>                                            | 16 (8 bits)                                                                          |
| <b>Cable Type</b>                                               | Proprietary standard                                                                 |
| <b>Cable Length</b>                                             | 1 meter                                                                              |

## Auxiliary Input and Output Characteristics

| Sync in/out                     |                                                       |
|---------------------------------|-------------------------------------------------------|
| Connector Type                  | Infiniband 4X connector on rear panel (custom pinout) |
| Master to Slave Delay (typical) | TBD                                                   |
| Marker Output                   |                                                       |
| Connectors                      | SMA on Front Panel                                    |
| Number of Connectors            | 1 2 4                                                 |
| Output Impedance                | 50 $\Omega$                                           |

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|                                                   |                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output level (into 50 <math>\Omega</math>)</b> |                                                                                                                                                                                                                                                                                                             |
| Voltage Window                                    | -0.5 V to 1.65 V                                                                                                                                                                                                                                                                                            |
| Amplitude                                         | 100 mVpp to 2.15 Vpp                                                                                                                                                                                                                                                                                        |
| Resolution                                        | 1 mV                                                                                                                                                                                                                                                                                                        |
| Accuracy                                          | $\pm(5\% \text{ setting} + 25 \text{ mV})$                                                                                                                                                                                                                                                                  |
| <b>Switching characteristics</b>                  |                                                                                                                                                                                                                                                                                                             |
| Max Update Rate (True Arb Mode)                   | 6.16 Gbps                                                                                                                                                                                                                                                                                                   |
| Max Data Rate (True Arb Mode)                     | >4 Gbps @ 1Vpp swing                                                                                                                                                                                                                                                                                        |
| Max Frequency (AFG Mode)                          | 96.5 MHz (continuous mode)                                                                                                                                                                                                                                                                                  |
| Rise/fall time (10% to 90%, 2 Vpp)                | <150 ps                                                                                                                                                                                                                                                                                                     |
| Jitter (rms)                                      | <10 ps                                                                                                                                                                                                                                                                                                      |
| <b>Marker out to analog channel skew</b>          |                                                                                                                                                                                                                                                                                                             |
| Range                                             | True Arb Mode: 0 to 2.3 $\mu$ s<br>AFG Mode: 0 to 100 sec. in Contin. Mode, 0 to 2.25 $\mu$ s in Trig. Mode                                                                                                                                                                                                 |
| Resolution                                        | True Arb Mode: 1/64 of DAC sampling period, AFG Mode: 5 ps                                                                                                                                                                                                                                                  |
| Accuracy                                          | $\pm(1\% \text{ of setting} + 5 \text{ ps})$                                                                                                                                                                                                                                                                |
| Initial skew                                      | < 20 ps                                                                                                                                                                                                                                                                                                     |
| <b>Trigger/Event Inputs</b>                       |                                                                                                                                                                                                                                                                                                             |
| Connector                                         | SMA on the Front Panel                                                                                                                                                                                                                                                                                      |
| Number of Trigger Inputs                          | 2 (Trig.in 1, Trig.in 2)                                                                                                                                                                                                                                                                                    |
| Input Impedance                                   | 50 $\Omega$ / 1 k $\Omega$                                                                                                                                                                                                                                                                                  |
| Slope/Polarity                                    | Positive or negative or both                                                                                                                                                                                                                                                                                |
| Input Damage Level                                | < -15 V or > +15 V                                                                                                                                                                                                                                                                                          |
| Threshold Control Level                           | -10 V to 10 V                                                                                                                                                                                                                                                                                               |
| Resolution                                        | 50 mv                                                                                                                                                                                                                                                                                                       |
| Threshold Control Accuracy                        | $\pm(10\% \text{ of }  \text{setting}  + 0.2 \text{ V})$                                                                                                                                                                                                                                                    |
| Input Voltage Swing                               | 0.5 Vp-p minimum                                                                                                                                                                                                                                                                                            |
| Minimum Pulse Width (1 Vp-p)                      | 3 ns                                                                                                                                                                                                                                                                                                        |
| Trigger/gate input to Analog Output Delay         | <b>Slow (synchronous) trigger</b><br>AFG mode: < 355 ns (< 405 ns in triggered sweep mode)<br>True Arb mode: <1550 * DAC clock period(ns) + 10 ns<br><br><b>Fast (asynchronous) trigger</b><br>AFG mode: < 335 ns (< 385 ns in triggered sweep mode)<br>True Arb mode: <1360 * DAC clock period(ns) + 27 ns |
| Trigger In to Output Jitter (rms)                 | AFG mode: < 20 ps<br>True Arb mode: 0.29*DAC clock period                                                                                                                                                                                                                                                   |
| Trigger In Programmable Delay Range               | 0 ps to 2418 ps                                                                                                                                                                                                                                                                                             |
| Trigger In Programmable Delay Resolution          | 78 ps                                                                                                                                                                                                                                                                                                       |
| Maximum Frequency                                 | AFG: 65 MTps on Rising/Falling Edge, 80 MTps on Both Edges<br>True Arb mode: 1/ (Period of the Analog Waveform + 48 DAC Clock period)<br>MTps = Mega Transitions per second                                                                                                                                 |
| <b>Reference clock input</b>                      |                                                                                                                                                                                                                                                                                                             |
| Connector Type                                    | SMA on Rear Panel                                                                                                                                                                                                                                                                                           |
| Input Impedance                                   | 50 $\Omega$ , AC coupled                                                                                                                                                                                                                                                                                    |
| Input Voltage Range                               | 0.2 Vpp to 2 Vpp                                                                                                                                                                                                                                                                                            |
| Damage Level                                      | Maximum Input voltage: -0.3 V to 3.6 V<br>Maximum input power: 30 dBm (50 $\Omega$ )                                                                                                                                                                                                                        |
| Frequency Range                                   | 5 MHz to 200 MHz                                                                                                                                                                                                                                                                                            |
| Frequency Resolution                              | 1 Hz                                                                                                                                                                                                                                                                                                        |



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| Reference clock output              |                                                                                                                                                                                                                   |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Connector Type                      | SMA on Rear Panel                                                                                                                                                                                                 |
| Output Impedance                    | 50 $\Omega$ , AC coupled                                                                                                                                                                                          |
| Frequency                           | 10 MHz TCXO   100 MHz VCOCXO (Optional)                                                                                                                                                                           |
| Initial Accuracy @ 25 °C            | $\pm 1.0$ ppm   $\pm 500$ ppb (Opt.)                                                                                                                                                                              |
| Aging                               | $\pm 1.0$ ppm/year   $\pm 500$ ppb/year (Opt.)                                                                                                                                                                    |
| Stability vs. Temperature           | $\pm 1$ ppm   $\pm 50$ ppb(Opt.)                                                                                                                                                                                  |
| Amplitude                           | 1.65 Vpp                                                                                                                                                                                                          |
| Phase Noise @ 20 MHz Carrier        | -120 dBc/Hz at 100 Hz ;<br>-140 dBc/Hz at 1 KHz;<br>-150 dBc/Hz at 10 KHz                                                                                                                                         |
| Phase Noise @ 100 MHz Carrier(Opt.) | -120 dBc/Hz at 100 Hz ;<br>-145 dBc/Hz at 1 KHz;<br>-150 dBc/Hz at 10 KHz                                                                                                                                         |
| External Clock Input                |                                                                                                                                                                                                                   |
| Connector Type                      | SMA on Rear Panel                                                                                                                                                                                                 |
| Input Impedance                     | 50 $\Omega$ , AC coupled                                                                                                                                                                                          |
| Frequency <sup>5</sup>              | True Arb: SampleRate / N where:<br>N = 4, 8, 16, 32 for SampleRate = 5.0 $\div$ 6.16 GHz<br>N = 2, 4, 8, 16, 32 for SampleRate = 3.08 $\div$ 5.0 GHz<br>AFG: 192.5 MHz, 385 MHz, 770 MHz or 1540 MHz (selectable) |
| Input Power Range                   | +0 dBm to +10 dBm                                                                                                                                                                                                 |
| Damage Level                        | 15 dBm                                                                                                                                                                                                            |
| Sync Clk Out                        |                                                                                                                                                                                                                   |
| Connector Type                      | SMA on Rear Panel                                                                                                                                                                                                 |
| Output Impedance                    | 50 $\Omega$ , AC coupled                                                                                                                                                                                          |
| Frequency                           | AFG Mode: 6.16GHz / N where N=16, 32, 64, ..., 2048<br>AWG Mode: 6.16GHz / 16 to 6.16GHz / 4096                                                                                                                   |
| Amplitude                           | 1 Vpp into 50 Ohm                                                                                                                                                                                                 |
| External Modulation input           |                                                                                                                                                                                                                   |
| Connector Type                      | SMA on Rear Panel                                                                                                                                                                                                 |
| Input Impedance                     | 10 K $\Omega$                                                                                                                                                                                                     |
| Number of Inputs                    | 1                                                                                                                                                                                                                 |
| Bandwidth                           | 10 MHz with 50 MS/s sampling rate                                                                                                                                                                                 |
| Input Voltage Range                 | -1 V to +1 V (except FSK, PSK).<br>FSK, PSK: 0 V $\div$ 3.3 V with 1.65V fixed threshold                                                                                                                          |
| Vertical Resolution                 | 12-bit                                                                                                                                                                                                            |
| Pattern Jump In (optional)          |                                                                                                                                                                                                                   |
| Connector Type                      | DSUB15                                                                                                                                                                                                            |
| Input Signals                       | DATA[0..7] + Data Select + Load                                                                                                                                                                                   |
| Internal Data Width                 | 14 bit, multiplexed using Data_Select                                                                                                                                                                             |
| Number of Addressable Entries       | 16384                                                                                                                                                                                                             |
| Data Rate                           | DC to 1 MHz                                                                                                                                                                                                       |
| Input Range                         | VIL = 0 V to 0.8 V / VIH= 2 V to 3.3 V                                                                                                                                                                            |
| Impedance                           | Internal 1 k $\Omega$ pull-up resistor to Vcc (3.3 V)                                                                                                                                                             |
| Power                               |                                                                                                                                                                                                                   |
| Source Voltage and Frequency        | 100 to 240 VAC $\pm 10\%$ @ 45-66 Hz                                                                                                                                                                              |

<sup>5</sup> When using the External Clock Input the SampleRate must be in the range 3.08  $\div$  6.16 GHz

# Model 685 Arbitrary Waveform Generator

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Max. Power Consumption</b>        | Max. 100W (Model 685-2C / 2CD)<br>Max. 200W (Model 685-4C / 4CD)<br>Max. 300W (Model 685-8C / 8CD)                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Environmental Characteristics</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Temperature (operating)</b>       | +41 °F to 104 °F [+5 °C to +40 °C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Temperature (non-operating)</b>   | -4 °F to 140 °F [-20 °C to +60 °C]                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Humidity (operating)</b>          | 5% to 80% relative humidity with a maximum wet bulb temperature of 84°F (29°C) at or below +104°F (40°C), (upper limit de-rates to 20.6% relative humidity at +104°F (40°C)). Non-condensing.                                                                                                                                                                                                                                                                                                                    |
| <b>Humidity (non-operating)</b>      | 5% to 95% relative humidity with a maximum wet bulb temperature of 104°F (40°C) at or below +140°F (60°C), upper limit de-rates to 29.8% relative humidity at +140°F (60°C). Non-condensing.                                                                                                                                                                                                                                                                                                                     |
| <b>Altitude (operating)</b>          | 9,842 feet (3,000 meters) maximum at or below 77°F (25°C)                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Altitude (non-operating)</b>      | 39,370 feet (12,000 meters) maximum                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>EMC and Safety</b>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Safety</b>                        | EN61010-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Main Standards</b>                | EN 61326-1:2013 – Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Immunity</b>                      | EN 61326-1:2013                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>System Specifications</b>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Display</b>                       | 7", 1024x600, capacitive touch LCD                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Operative System</b>              | Windows 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>External Dimensions</b>           | W 17.6 in – H 5.4 in – D 12.6 in (3U 19" rackmount)<br>(445 mm – 135 mm – 320 mm)                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Weight</b>                        | Max. 26.45 lbs (12 Kg)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Front panel connectors</b>        | CH N OUTPUT (SMA) where N=2,4,8 depending on the model<br>MARKER N OUT (SMA) where N=1,2,4 depending on the model<br>TRG IN N(SMA) where N =1,2<br>2 USB 3.0 ports                                                                                                                                                                                                                                                                                                                                               |
| <b>Rear panel connectors</b>         | Ref. Clk. IN (SMA)<br>Ref. Clk. Out (SMA)<br>Ext. Mod. IN (SMA)<br>Sync Clk Out (SMA)<br>Ext Clk IN (SMA)<br>Sync IN (Infiniband 4X)<br>Sync OUT (Infiniband 4X)<br>Pattern Jump In (DSUB15) (685-XC-FSS opt. only)<br>POD X[7..0] where X=A,B,C,D depending on the model (Customized Mini SAS HD)<br>External Monitor ports (one or more)<br>2 USB 2.0 ports or more<br>4 USB 3.0 ports<br>Ethernet port (10/100/1000BaseT Ethernet, RJ45 port)<br>2 PS/2 keyboard and mouse ports<br>2 DPI ports<br>1 DVI port |
| <b>Hard Disk</b>                     | 1 TB SSD or better                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Processor</b>                     | Intel® Pentium 3.7 GHz (or better)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Processor Memory</b>              | 32 GB or better                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |